

Where Models Miss: A Cross-Platform Analysis of GUI Action Grounding

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Abstract

Large vision-language models (VLMs) achieve strong GUI grounding but require dedicated accelerators, making real-time deployment impractical. Small models (3B–7B) offer a practical alternative, yet their failure modes and capability limits on GUI grounding tasks remain poorly understood. We present Click2Act, an open evaluation framework for GUI grounding, and use it to systematically evaluate four small pure-vision VLMs AGUVIS-7B, Holo2-4B, UI-AGILE-3B, and AGUVIS-7B augmented with UI-Zoomer across two benchmarks: ScreenSpot-Pro for desktop grounding and AndroidControl for mobile action prediction. Our analysis reveals that coordinate precision, not action type classification, is the universal bottleneck at this scale: small models reach at most 40.5ScreenSpot-Pro compared to over 70augmentation via UI-Zoomer improves AGUVIS-7B by 3.84 percentage points without any additional training, suggesting that adaptive zoom is a viable direction for bridging the accuracy gap without scaling model size.

1. Introduction

The Graphical User Interface (GUI) is the gateway for human interaction in any application, so the need for GUI agents has gained increasing attention in recent years. Recently, GUI agents have been powered by Vision Language Models (VLMs), enabling stronger screen comprehension and accurate element localization. Large models such as **Holo2-235B** (Company, 2025), **UI-Venus-72B** (Venus-Team et al., 2026), and **UI-TARS-72B** (Qin et al., 2025) now score over 75

Prior evaluation efforts have focused on establishing state-of-the-art performance, primarily analyzing large-model regimes. **VenusBench-GD** (Zhou et al., 2025) evaluates 20+ models but centers its analysis on large frontier models.

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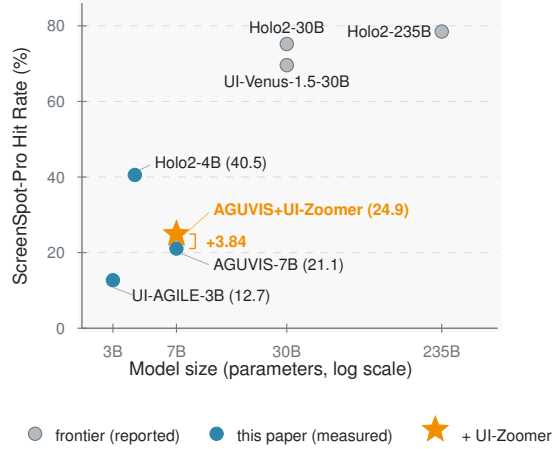


Figure 1. **The small-model accuracy gap on ScreenSpot-Pro.** Frontier models (gray circles, 72B–235B) clear 70%. The small open models we study (colored stars, 3B–7B) leave 35–65 points of headroom; even the best, Holo2-4B, lags the frontier by ~35 points. Click2Act dissects where this gap originates. Frontier numbers are illustrative; see Table 2 for measured small-model values.

Similarly, **UGround** (Gou et al., 2025) demonstrates universal grounding capabilities across 7B to 72B scales, yet no systematic study examines the 3B to 7B regime. Where exactly small models fail, whether their error modes differ qualitatively from those of large models, and whether inference-time strategies can compensate for their reduced capacity remain open questions.

In this paper, we present Click2Act¹, an open evaluation framework for GUI grounding, and use it to conduct a systematic evaluation of four small GUI agents: **AGUVIS-7B** (Xu et al., 2024), **Holo2-4B** (Company, 2025), **UI-Agile-3B** (Lian et al., 2025), **AGUVIS + UI-Zoomer** AGUVIS-7B augmented with UI-Zoomer (Tang et al., 2026). We evaluate across two domains, the desktop domain using **ScreenSpot-Pro** (Li et al., 2025) and the mobile domain using **AndroidControl** (Li et al., 2024), providing per-model failure taxonomies, efficiency profiling, and a study of the effect of the UI-Zoomer component.

Our contributions are: (1) Click2Act, an extensible open-

¹<https://github.com/YaraHisham61/Click2Act>

source framework supporting pluggable agents, benchmarks, and YAML-based configuration; (2) the first in-depth analysis of the small-model (3B–7B) regime for GUI grounding; (3) a quantitative efficiency-accuracy tradeoff analysis that clarifies what users sacrifice by choosing small models; and (4) an evaluation of UI-Zoomer as an inference-time augmentation, assessing whether adaptive zoom can help bridge the performance gap for small models.

2. Related Work

Multimodal large language models (MLLMs) are often the base model behind GUI agents, and benchmarks now focus more on multi-step agent tasks than on single-turn question answering. GUI grounding means mapping language to the right on-screen control. Recent work compares grounding and actions on desktop and mobile, not just in small lab demos.

2.1. Evolution of GUI agent architectures

Much of the literature groups progress into three lines: supervised fine-tuning (SFT), reinforcement learning (RL), and light-weight changes at inference time.

Supervised fine-tuning. AGUVIS-7B (Xu et al., 2024) uses two phases: planning with a structured inner monologue, then grounding those steps into actions. OS-Atlas (Wu et al., 2024), SeeClick (Cheng et al., 2024), and UGround (Gou et al., 2025) train for shared visual grounding so one model can cover many apps and OS layouts.

Reinforcement learning. Holo2-4B (Company, 2025) adds an RL stage with Group Relative Policy Optimization (GRPO) to tighten multi-step navigation. UI-Agile-3B (Lian et al., 2025) uses a smooth reward for near-miss clicks and crop-based resampling that zooms the screenshot so hard training pairs stay useful without increasing the network.

Inference-time augmentation. UI-Zoomer (Tang et al., 2026) runs an extra, zoomed forward pass when the model is unsure where to click, in order to achieve sharper boxes on dense screens without retraining the base weights.

2.2. Benchmarking and saturation

The evaluation of these agents reveals a historical divergence between mobile and desktop research. Earlier benchmarks often specialized in a single platform, such as Android Control (Li et al., 2024) for mobile GUI datasets. This specialization limits applicability to broader user interactive scenarios and fails to reflect the cross-platform generalization needed for everyday user tasks. Our study addresses this literature gap by conducting a cross-platform analysis that integrates evaluations across desktop and mobile environments.

Furthermore, as models achieve high accuracy on standard datasets, there is a shifting need toward more challenging environments. ScreenSpot-Pro (Li et al., 2025) targets professional desktop applications where tool-rich, high resolution screenshots provide a strong discriminator between models.

3. Method

Click2Act is an open, extendable evaluation framework for GUI agents. It supports two evaluation modes—point grounding and action prediction—and is designed to be modular: new agents and benchmarks are registered independently and paired at runtime via YAML configuration files as illustrated in Figure 2.

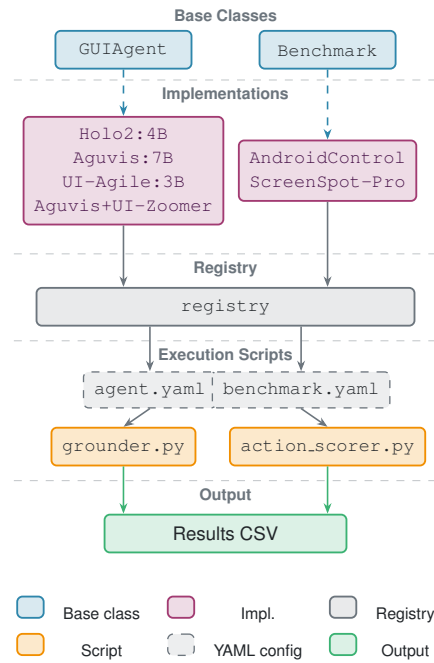


Figure 2. **Click2Act framework overview.** Base classes (GUIAgent, Benchmark) are dynamically subclassed via a registry. Scripts load YAML configs to run inference and log CSV results. Legend: dashed (inheritance), solid (data flow).

3.1. Evaluated Models

We evaluate four small pure-vision VLMs, all used with no additional fine-tuning. Table 1 summarizes each model.

All evaluated models fall in the 3B–7B parameter range, targeting practical deployment on consumer devices like laptops and workstations without cloud infrastructure. Despite this appeal, how much accuracy they sacrifice compared to flagships such as UI-Venus-1.5-30B-A3B (Venus-Team et al., 2026) and Holo2-235B-A22B (Company, 2025), and where exactly they fail, remains an open question.

Table 1. Models evaluated in this study.

Model	Params
UI-AGILE-3B	3B
Holo2-4B	4B
AGUVIS-7B	7B
AGUVIS+UI-Zoomer	7B

AGUVIS-7B (Xu et al., 2024) is a fully open-source autonomous GUI agent built on Qwen2-VL that supports all platforms. It operates in two stages: (1) a planning stage where the model generates a structured *inner monologue*—consisting of a textual description of the current screen, internal reasoning toward the high-level goal, and low-level action instructions—and (2) a grounding stage where the inner monologue is translated into executable actions. It was one of the first small fully open-source vision-based GUI agents to demonstrate competitive performance on ScreenSpot.

Holo2-4B (Company, 2025) is a fully open-source autonomous GUI agent built on Qwen3-VL that supports all platforms. It is trained in two stages: (1) a large-scale supervised fine-tuning (SFT) stage on screen localization and element grounding, and (2) an online reinforcement learning stage using Group Relative Policy Optimization (GRPO) to refine its multi-step reasoning and navigation policies.

UI-Agile-3B (Lian et al., 2025) is a fully open-source GUI grounding agent built on Qwen2.5-VL-3B. It is aiming at improving both the reinforcement fine tuning and the inference stages of GUI agents. their contribution are in three main points: firstly, UI-Agile applies simple thinking strategy that encourages thoughts within an "ideal range" of length, secondly, UI-Agile provides a continuous reward function that treats a click on the very edge of a button the same as a click in the center, thirdly, UI-Agile applies a cropping-based resampling strategy that dynamically adjusts the difficulty by cropping (zooming in) on the original screenshot.

AGUVIS + UI-Zoomer. We pair AGUVIS with UI-Zoomer (Tang et al., 2026) to investigate whether inference-time augmentation alone can close the accuracy gap with larger models, without increasing model size or requiring additional training. UI-Zoomer runs in two stages. In the first stage, the model samples N candidate coordinate predictions and computes a confidence score $c = C_{\text{tokens}} + C_{\text{spatial}}$. The token confidence averages the geometric-mean token probability across samples:

$$C_{\text{tokens}} = \frac{1}{N} \sum_{j=1}^N \exp\left(\frac{1}{L_j} \sum_{i=1}^{L_j} \log p_i^{(j)}\right)$$

where L_j is the number of non-template tokens in sample

j . The original UI-Zoomer defines spatial consensus over bounding boxes as the mean pairwise IoU:

$$C_{\text{spatial}} = \frac{1}{N(N-1)} \sum_{i \neq j} \text{IoU}(\mathbf{b}_i, \mathbf{b}_j)$$

Since AGUVIS produces point predictions rather than bounding boxes, we reformulate spatial consensus by treating each predicted point $\mathbf{p}_j$ as the mean of an isotropic Gaussian with standard deviation $\sigma = r \cdot \max(W, H)$, where $r = 0.02$. Under this assumption, the overlap between two point predictions is naturally captured by the joint probability $P(\mathbf{p}_i, \mathbf{p}_j)$, which yields a Gaussian kernel as an analog to IoU:

$$C_{\text{spatial}} = \frac{1}{N(N-1)} \sum_{i \neq j} \exp\left(-\frac{|\mathbf{p}_i - \mathbf{p}_j|^2}{2\sigma^2}\right)$$

The best candidate is then selected via consensus voting: each candidate accumulates votes from peers whose predictions are spatially close, and the candidate with the most support is chosen, with token confidence as a tiebreaker. In the original UI-Zoomer, a vote is cast when $\text{IoU}(\mathbf{b}_i, \mathbf{b}_j) > 0.5$; we replace this with a distance threshold under our Gaussian formulation, giving $v_i = \sum_{j \neq i} \mathbb{I}[|\mathbf{p}_i - \mathbf{p}_j| < \sigma]$, and select $i^* = \arg \max_i (v_i, c_i)$.

If $c \geq \tau$ (we use $\tau = 0.65$), the voted point is returned directly. Otherwise, the second stage applies adaptive cropping: outlier points are removed by retaining the closest 75% to the centroid, and a square crop of side $s = \max(2\gamma\sigma_x, 2\gamma\sigma_y, m)$ is centered on the filtered mean, where $\gamma = 2.5$ is a scale factor and $m = 256$ is the minimum crop side. The crop is shifted rather than clipped to remain within image bounds. The model then re-predicts on the cropped region, and the output coordinate is mapped back via $x_{\text{pred}} = x_{\text{crop}} \cdot w_c + x_0$.

3.2. Experimented Benchmarks

ScreenSpot-Pro (Li et al., 2025) is a static GUI grounding benchmark targeting high-resolution desktop screenshots, with 1,580 samples distributed across 26 professional applications on Linux, macOS, and Windows. State-of-the-art models have not yet saturated its leaderboard, making it a strong discriminator between models. Each sample pairs a screenshot with a low-level instruction and a ground-truth bounding box. Where, the text targets outnumber icon targets in most applications, with the gap being especially large in domain-specific tools such as `matlab`, `solidworks` and `word`.

Android Control (Li et al., 2024) is a mobile GUI dataset. We evaluate on a subset of episodes from its test set, where each episode consists of a high-level goal and multiple steps. Each step pairs a screenshot and a low-level instruction

with a ground-truth action (action type, coordinate, text, and scroll direction).

3.3. Evaluation Metrics

Hit Rate (ScreenSpot-Pro) measures the fraction of samples for which the predicted point falls within the ground-truth bounding box.

Step Accuracy (Android Control) measures the fraction of steps for which the predicted action matches the ground truth on all applicable components. Coordinates are normalized to $[0, 1]$ by the screen dimensions, so a predicted coordinate (x', y') is considered correct if $\|(x', y') - (x^*, y^*)\|_2 \leq \tau$, where (x^*, y^*) is the ground-truth coordinate. We report results at $\tau \in \{0, 0.01, 0.05, 0.1\}$ to characterize each model’s coordinate precision across a range of tolerance levels.

Component-level accuracy decomposes step accuracy into component scores (action type, coordinate, text, direction), enabling identification of which prediction components are responsible for failures.

4. Experiments

All experiments of AGUVIS and UI-Agile were conducted on Kaggle using T4x2 GPUs. Experiments of HOLO2 were conducted on Modal using NVIDIA A10G 24G GPUs. All models were evaluated with no additional fine-tuning, using the inference configurations specified in their respective YAML files. Images entered into the models were preprocessed using smart resize.

4.1. ScreenSpot-Pro Results

We used hit rate as our evaluation metric (shown in Table 2), defined as the success rate of the model when its prediction falls within the bounding box coordinates.

Table 2. ScreenSpot Pro hit rate

Model	Accuracy (%)
UI-AGILE-3B	12.71
AGUVIS-7B	21.07
AGUVIS + UI-Zoomer	24.91
Holo2-4B	40.54

Table 3 reports the effect of 4-bit quantization on UI-AGILE-3B, evaluated on the first 78 samples of ScreenSpot-Pro. Applying INT4 quantization via `bitsandbytes` reduces accuracy from 7.69% to 5.13%, a drop of 2.56 percentage points, while providing no measurable reduction in inference latency. The bottleneck in UI-AGILE-3B lies in activation memory rather than weight memory: quantizing weights to 4-bit reduces the model size on disk, but pro-

vides no improvement in inference latency when processing images.

Table 3. Effect of 4-bit quantization on UI-AGILE-3B accuracy (%) on the first 78 samples of ScreenSpot Pro.

Configuration	Accuracy (%)
UI-AGILE-3B (BF16)	7.69
UI-AGILE-3B (INT4)	5.13

4.2. AndroidControl Results:

We evaluate two agents on the AndroidControl dataset. Predictions are made in normalized coordinates $[0, 1]^2$, and clicks are judged correct if the predicted point falls within Euclidean distance τ of the ground-truth target. Action type, typed text, and scroll direction are checked separately where applicable. Steps with `wait` or `open_app` ground-truth actions are excluded from evaluation (373 records), as the models either lack temporal context for wait decisions or predict `click` in place of `open_app`.

Table 4. Hit rate (%) on AndroidControl. Sample sizes differ between models.

Model	Hit Rate (%)	Samples
AGUVIS-7B	75.00	2988
Holo2-4B	58.40	5309

As shown in Table 4, AGUVIS-7B achieves a hit rate of 75.0% compared to 58.4% for HOLO2-4B at $\tau=0.1$. To further examine localization precision, Table 5 reports accuracy across stricter thresholds.

Table 5. Accuracy at varying coordinate thresholds δ .

δ	HOLO2-4B	AGUVIS-7B
0.10	58.4%	75.0%
0.05	52.6%	72.1%
0.01	37.7%	61.9%
0.00	7.4%	21.9%

4.3. Aguis + UI-Zoomer Results:

integrating UI-Zoomer achieves a 24.91% overall hit rate on ScreenSpot-Pro, improving the AGUVIS-7B baseline by +3.84 points. We break down performance into two stages based on voting confidence. In Stage-1, 42.7% of samples meet the confidence threshold ($c \geq \tau$) and bypass adaptive cropping. For these, the initial prediction hits the target 38.0% of the time. In Stage-2, the remaining 57.3% of samples fall below the threshold ($c < \tau$) and trigger

adaptive cropping. The crop isolates the target in 74.8% of these Stage-2 cases, setting an upper bound for recovery.

Notably, the combined probability of triggering a crop and successfully framing the target ($0.573 \times 0.748 \approx 0.4286$) reveals that in nearly 42.9% of all test cases, the system visually isolates the target but fails to act on it. This suggests a failure in task comprehension rather than visual grounding.

However, within this successfully cropped subset, the model correctly re-predicts the target only 18.8% of the time. This yields a 15.1% hit rate across all Stage-2 samples. We hypothesize this may stem from a language mismatch, as ScreenSpot-Pro contains Chinese instructions while AGU-VIS is primarily trained on English data. Though unconfirmed in this study, this potential linguistic bottleneck presents a clear direction for future investigation.

Table 6. Performance decomposition of the UI-Zoomer pipeline on ScreenSpot-Pro.

Pipeline Stage	
Stage-1: High-Confidence Short-Circuit ($c \geq \tau$)	42.7
Voted point hits target	38.0
Stage-2: Adaptive Crop & Re-Predict ($c < \tau$)	57.3
Target contained inside crop (recoverable)	74.8
Target recovered (re-predicted in GT bbox)	18.8
Final hit rate across Stage-2 samples	15.1
Overall Pipeline Hit Rate	24.91

5. Conclusion

We present Click2Act, an evaluation framework for GUI grounding, and use it to conduct a systematic analysis of four small VLMs (3B–7B) across two benchmarks: ScreenSpot-Pro for desktop grounding and AndroidControl for mobile action prediction. Across both benchmarks, coordinate precision emerges as the universal bottleneck: action type classification is largely solved at this scale, while fine-grained localization remains the primary source of failure. UI-Zoomer partially addresses this gap as a training-free inference-time augmentation, lifting AGU-VIS-7B from 21.07% to 24.91% on ScreenSpot-Pro without any additional training, suggesting that adaptive zoom is a promising direction for improving localization in small models. Future work includes fine-tuning small models on domain-specific data to close the desktop grounding gap, combining grounding agents with parsing-based approaches such as OmniParser for richer UI representations, and extending evaluation to multi-step task completion metrics beyond per-step accuracy.

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